



2.1.3 What you will learn

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This week introduces two concepts that have theoretical and practical importance: unitary matrices and the Singular Value Decomposition (SVD).

Upon completion of this week, you should be able to

- Determine whether vectors are orthogonal.
- Compute the component of a vector in the direction of another vector.
- Relate sets of orthogonal vectors to orthogonal and unitary matrices.
- Connect unitary matrices to the changing of orthonormal basis.
- Identify transformations that can be represented by unitary matrices.
- Prove that multiplying with unitary matrices does not amplify relative error.
- Use norms to quantify the conditioning of solving linear systems.
- Prove and interpret the Singular Value Decomposition.
- Link the reduced Singular Value Decomposition to the rank of the matrix and determine the best rank-k approximation to a matrix.
- Determine whether a matrix is close to being nonsingular by relating the Singular Value Decomposition to the condition number.

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